

CONTENTS

The whole system
 Tenancy
 Getting data in
 The ownership split
 The arithmetic rule
 Data to a board
 Themes and artwork
 Credentials and roles
 What runs where
 Guards
 Capabilities

SYSTEM MAP • SEPTEMBER 2026

Scoreboard, from source to wall

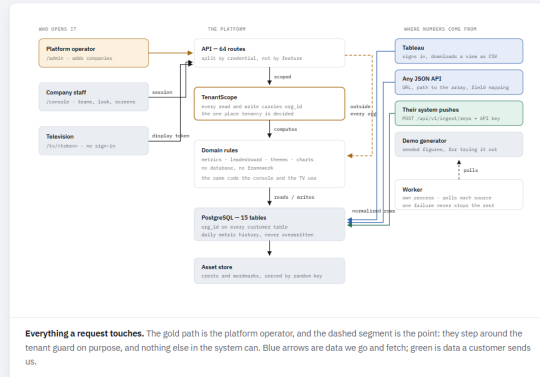
A multi-tenant sales leaderboard. Numbers arrive from a customer's reporting system; the platform owns the teams, the look, and the screens on the office televisions. This is how the parts fit together, and which of them exist to stop something going wrong.

API ROUTES	TABLES	CONNECTORS	BOARD NODES	CHART TYPES	TESTS
64	15	4	4	6	122

OVERVIEW

The whole system

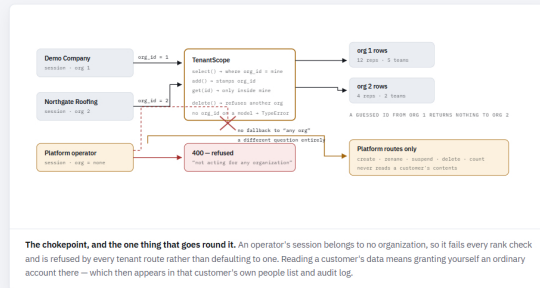
Three kinds of visitor, three kinds of credential, one API, and one place where the question “which customer is this?” is answered. Data reaches the database from two directions: systems we can reach, and systems that reach us.



ISOLATION

One customer cannot see another

Every table holding customer data carries `org_id`, and every query goes through `TenantScope` — a session bound to one organization. It is not a convenience wrapper. It is the single place where tenancy is applied, so isolation is a property of the architecture rather than of each developer remembering a `WHERE` clause.

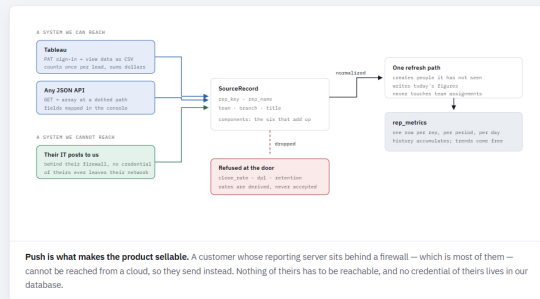


INGESTION

Two directions, one normalized row

Whether figures arrive from a Tableau view, an arbitrary JSON endpoint, or a customer's own script, they become the same `SourceRecord` before anything else sees them. Adding a customer's system means adding one module behind that interface; nothing downstream changes.

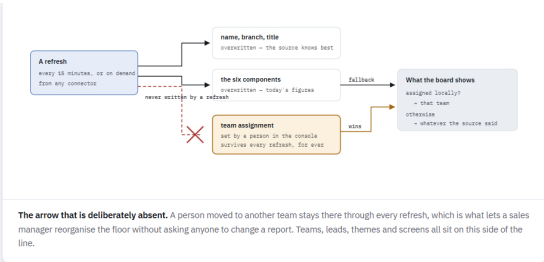
tableau — we fetch test — we fetch ingest — they send mock — generated



THE FIRST RULE

The source supplies numbers. The platform owns the organization.

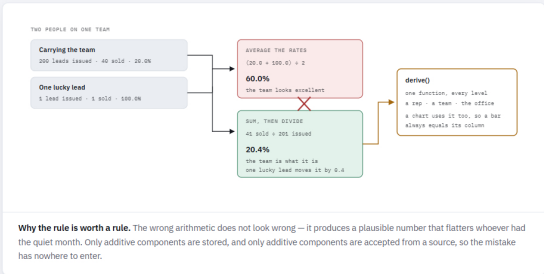
This is the rule the original product existed for. Tableau knows what a salesperson sold; it does not know that the sales manager moved them to a different team last Tuesday. So a refresh writes figures and identity, and is structurally unable to write a team assignment.



THE SECOND RULE

Rates are calculated, never averaged

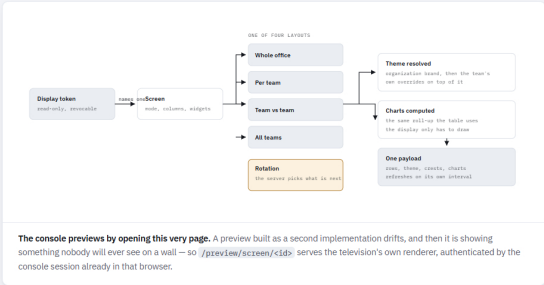
Six values add up and are stored: issued leads, pitched leads, sold leads, gross, pending and net. Everything else — close rate, pitched rate, DPL, retention, average sale — is recalculated from those components at whatever level is being totalled. There is exactly one implementation, so a team total cannot disagree with the rows above it.



RENDERING

From stored figures to a wall

A television asks once and draws. One request returns the ranked rows, the resolved theme, every team's crest and colour, and each chart already computed — because every extra request is another thing that can be half-answered on a wall nobody is watching.



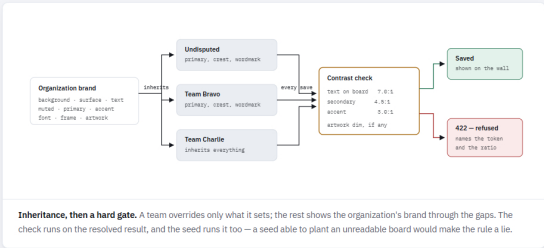
big number bar donut trend gauge leader list

Six chart types, not "any chart you like". The long list produces boards nobody can read from the far side of a sales floor. A donut refuses a rate, because a slice has to be a share of a whole; a donut trimmed to the top few names the remainder rather than dropping it.

IDENTITY

Every team can carry its own look

Customers get colours, artwork and a font from a curated set. They never get CSS. A television runs unattended for months in a room where nobody can fix it, so a theme that would be unreadable is **refused on save**, not saved with a warning.



Artwork has three slots rather than one logo, because on a wall a team is recognised by its crest long before its name is read: a **hero** banner across the top, a **badge** beside each person in the team column, and a modest **header logo** for boards where a hero is too much. Uploads are identified by their own first bytes, not by what they claim to be, and SVG is refused outright because it can carry script.

ACCESS

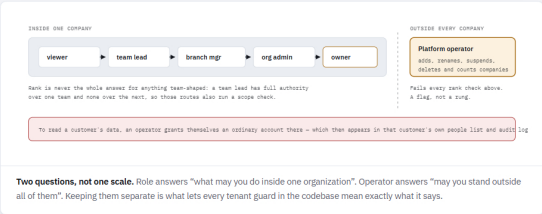
Three credentials, two ladders

The three credential types are deliberately not interchangeable, and the two kinds of authority are deliberately not the same ladder. Operator status is a flag on the account, not a very senior role — folding it into the rank would quietly turn every permission check in the system into "...or the operator", and no reader of those call sites would know.

CREDENTIAL	CARRIED BY	CAN DO	ENDS WHEN
ss_... session	a person in the console	everything their role allows, inside one company	sign-out, expiry, role change, removal, suspension
sb_... API key	a machine pushing data	write to the ingest endpoint, nothing else	revoked in the console

tv_... display link	a television	read one screen	revoked, or replaced by a new link
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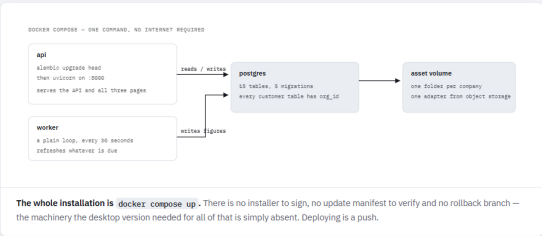
Sessions are rows, not self-contained tokens. When someone leaves a company the customer expects access to stop immediately, and a JWT cannot be withdrawn before it expires. A suspended company is checked on every request as well as at sign-in – a suspension that lasts until someone presses the login button again is not a suspension.



OPERATIONS

What runs where

Four processes and two volumes. The refresh loop is its own process rather than a thread inside the API, so a slow or hanging source cannot make the leaderboard stop answering: the API serves reads, the worker does the waiting.



SAFETY NET

The tests that guard the shape

Most of the 122 tests check behaviour. Four check that the system keeps the shape it was given, because those properties are the kind that decay quietly between one change and the next.

GUARD	FAILS WHEN	WHY IT EXISTS
every table carries <code>org_id</code>	a new table is added without it	a table without it is cross-tenant readable and nobody would notice
every write needs a credential	a mutating route is added with no auth dependency	the mistake that actually happens; it once passed while inspecting nothing
platform routes need the operator flag	a platform route is added without it	the one module that crosses tenants must not grow an unguarded door
models match migrations	a model changes with no migration	invisible in development, surfaces as a broken deploy

A GUARD THAT GUARDED NOTHING

The route check passed on its first run because current FastAPI does not flatten included routers into `app.routes` - the loop saw only the documentation endpoints. It now walks the tree, asserts it can see the application at all, and was verified by planting an unguarded endpoint to confirm it catches one. A green test that tests nothing is worse than no test.

REFERENCE

Everything it can do today

AREA	CAPABILITIES
Platform	Add a company with its first owner account · rename · suspend and reactivate · delete with typed confirmation · counts per company · grant yourself support access
Company	Branches · teams with leads and roles · move people between teams · delete a team only once every member has a destination · accounts, roles and scopes · password change and admin reset · audit log
Data	Tableau · any JSON API with field discovery and mapping · push endpoint with API keys · demo generator · scheduled refresh per source · connection test · encrypted credentials, never returned
Boards	Four modes · six chart types · chosen columns · ranking metric · rotation across screens · preview that is the board itself · revocable and re-issuable links
Look	Six colour tokens · six fonts · frame styles · hero, badge, logo and background artwork · per-team overrides inheriting the company brand · contrast enforced on save
Console	Home dashboard with office totals and team comparison charts · a page per team with its own statistics · live board · people · screens and widget builder · displays and rotation

Not built yet

Mail delivery - so an administrator creates an account and hands over a first password rather than sending an invitation, and resets one the same way. Object storage, though uploads already sit behind an interface one adapter away from it. Per-company time zones; everything is UTC. Billing. Rate limiting counts per process, which is exact with one API container and weaker, though never wrong, with several.